TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

SOCH COLLEGE OF IT

LAB REPORT OF

“SOFTWARE ENGINEERING”



SUBMITTED TO:

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SUBMITTED BY

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Define software engineering. Explain the roles of software engineering?

Ans: Software engineering is the technological and managerial discipline concerned with systematic production and maintenance of software products that are developed and modified on time and within cost estimates.

The roles of software engineer are as follows:

i. Should be a good programmer, be well-versed in data structures and algorithms, and be fluent in one or more programming languages.
ii. Should be familiar with several design approaches, be able to translate vague requirements and desire into precise specifications and be able to converse with the use of a system in terms of application.
iii. Needs the ability to move among several levels of abstraction at different stages of the projects, from specific application procedures and requirements, to abstraction for software systems, to a specific design for system and finally to the detailed coding level.
iv. Should be able to build and use a model of the application to guide choices of the many trade offs that he or she will face. The model id used to answer questions about both the behaviors of the system and its performance.
v. Needs communication skills and interpersonal skills. He also needs the ability to schedule work both of his own and that of others.
vi. Some primary goals of software engineering are:
    - To improve the quality of the software products.
    - To increase the productivity and
    - To give job satisfactions to the software engineers.

#compare Waterfall model with incremental model.

Ans:

S.N	Key	Incremental Model	Waterfall Model
1.	Defination	Incremental model is the development modeling which the entire model is divided into various sub development phase where corresponding testing for each development phase is practices.	Waterfall model there is first development of the application and after which the different testing of application takes place.
2.	Types/ Natures	As mentioned above that in incremental model the execution of the phases i.e. development and testing happen in a sequential manner so types of incremental is sequential/parallel in nature	WaterFall model is a relatively linear sequential design approach as each phase should be completed in order to reach the next phase. So type of this model is continuous in nature.
3.	Testing And validation	In this model each development phase get tested at its own level and hence no pending testing occurs in this model also if any validation requires to be implemented then it could be implemented at that phases.	In this model the testing occurs after development is completed and thus if any missing validation is identified to be implemented then first that phase of development needs to be recognized and then that validation get implemented.
4.	Cost and complexity	As sequential phases need to be functional hence the cost is higher as compared to that of waterfall model also the complexity is more than waterfall.	Here due to linear development only one phases of development is operational and hence cost and complexity is low as compared to incremental model
5.	Defects	Here probability of total number of defects in this development of application is low as testing is done in parallel to the development.	But here the probability of total number of defects in the development of application is high as testing is done post development.

#Explain any two iterative process model.

Ans: The iterative life cycle model comprises repeating the below-mentioned four stages as a sequence. These are:

1. Requirements Phase: In the requirements phase of software development, the system related information is gathered and analyzed. The collected requirements are then planned accordingly for developing the system.
2. Design Phase: In the Design phase, the software solution is prepared to meet the necessities for the design. The system design may be a new one or the extension of a previous build one.
3. Implementation and Test: In the implementation as well as a test phase, the system is developed by coding and building the user interface and modules which is then incorporated and tested.
4. Review Phase: The review phase is where the software is estimated and checked as per the current requirement. Then, further requirements are reviewed discussed and reviewed to propose for an update in the next iteration.

At every stage and cycle of this model, the decision is taken if the system created by the iteration or cycle will be rejected or not, or needs to process further for next cycle which is technically termed as incremental prototyping. Ultimately a situation will come, in which by implementing this model, the system requirements get complete & then the software can be distributed or delivered to the customer.

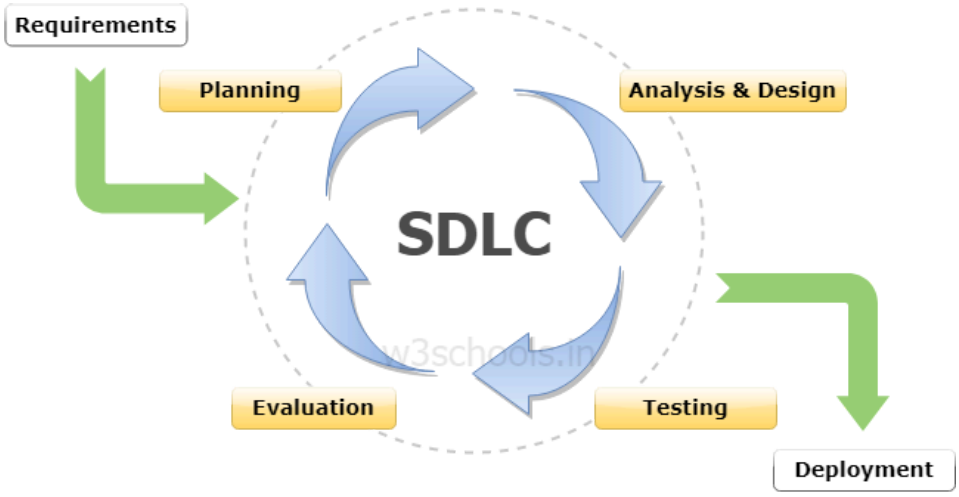


Fig: SDLC Iterative Model

#Explain any two Evolutionary process model.
Ans: Evolutionary models are iterative type models that allow to develop more complete versions of the software.
Following are the evolutionary process models.
1. The prototyping model
2. The spiral model
3. Concurrent development model

1. The Prototyping model

- Prototype is defined as first or preliminary form using which other forms are copied or derived.
- Prototype model is a set of general objectives for software.
- It does not identify the requirements like detailed input, output.
- It is software working model of limited functionality.
- In this model, working programs are quickly produced.

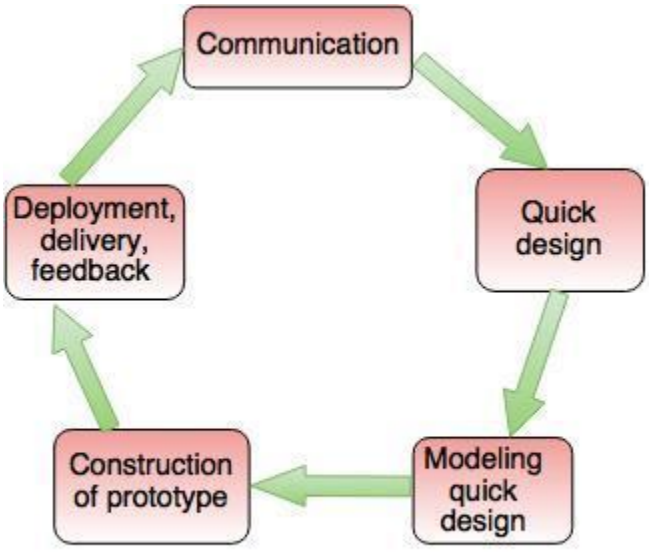


Fig. - The Prototyping Model

The different phases of Prototyping model are:

1. Communication
In this phase, developer and customer meet and discuss the overall objectives of the software.

2. Quick design

- Quick design is implemented when requirements are known.
- It includes only the important aspects like input and output format of the software.
- It focuses on those aspects which are visible to the user rather than the detailed plan.
- It helps to construct a prototype.

3. Modeling quick design

- This phase gives the clear idea about the development of software because the software is now built.
- It allows the developer to better understand the exact requirements.

4. Construction of prototype

The prototype is evaluated by the customer itself.

5. Deployment, delivery, feedback

- If the user is not satisfied with current prototype then it refines according to the requirements of the user.
- The process of refining the prototype is repeated until all the requirements of users are met.
- When the users are satisfied with the developed prototype then the system is developed on the basis of final prototype.

Advantages of Prototyping Model

- Prototype model need not know the detailed input, output, processes, adaptability of operating system and full machine interaction.
- In the development process of this model users are actively involved.
- The development process is the best platform to understand the system by the user.
- Errors are detected much earlier.
- Gives quick user feedback for better solutions.
- It identifies the missing functionality easily. It also identifies the confusing or difficult functions.

Disadvantages of Prototyping Model:

- The client involvement is more and it is not always considered by the developer.
- It is a slow process because it takes more time for development.

- Many changes can disturb the rhythm of the development team.
- It is a thrown away prototype when the users are confused with it.

2. The Spiral model

- Spiral model is a risk driven process model.
- It is used for generating the software projects.
- In spiral model, an alternate solution is provided if the risk is found in the risk analysis, then alternate solutions are suggested and implemented.
- It is a combination of prototype and sequential model or waterfall model.
- In one iteration all activities are done, for large project's the output is small.

The framework activities of the spiral model are as shown in the following figure.

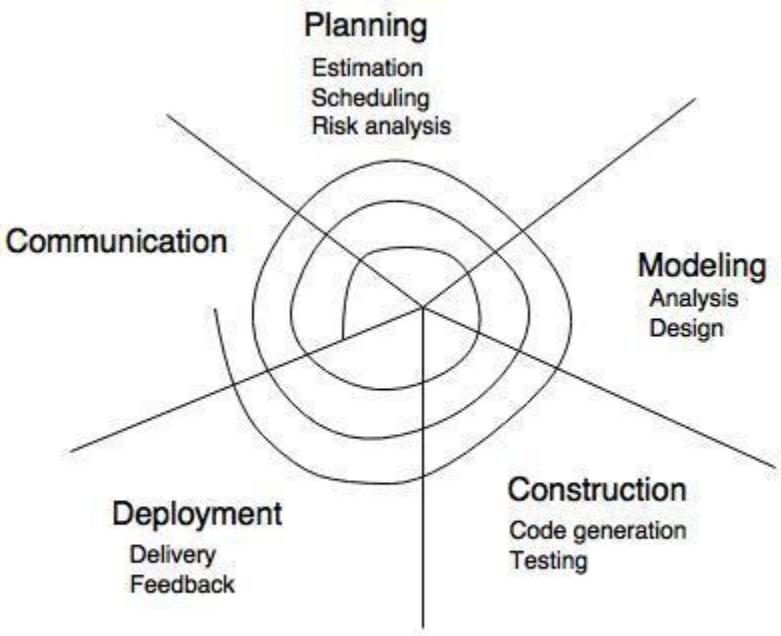


Fig. - The Spiral Model

NOTE: The description of the phases of the spiral model is same as that of the process model.

Advantages of Spiral Model

- It reduces high amount of risk.
- It is good for large and critical projects.
- It gives strong approval and documentation control.
- In spiral model, the software is produced early in the life cycle process.

Disadvantages of Spiral Model

- It can be costly to develop a software model.
- It is not used for small projects.

3. The concurrent development model

- The concurrent development model is called as concurrent model.
- The communication activity has completed in the first iteration and exits in the awaiting changes state.
- The modeling activity completed its initial communication and then go to the underdevelopment state.
- If the customer specifies the change in the requirement, then the modeling activity moves from the under development state into the awaiting change state.
- The concurrent process model activities moving from one state to another state.

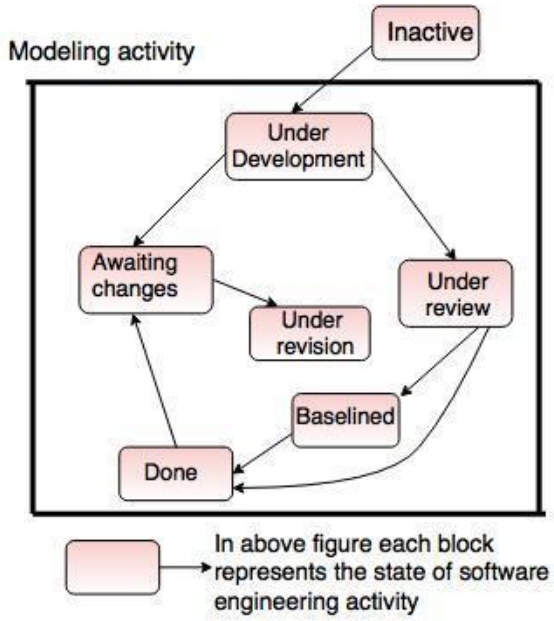


Fig. - One element of the concurrent process model

Advantages of the concurrent development model

- This model is applicable to all types of software development processes.
- It is easy for understanding and use.
- It gives immediate feedback from testing.
- It provides an accurate picture of the current state of a project.

Disadvantages of the concurrent development model

- It needs better communication between the team members. This may not be achieved all the time.
- It requires to remember the status of the different activities.